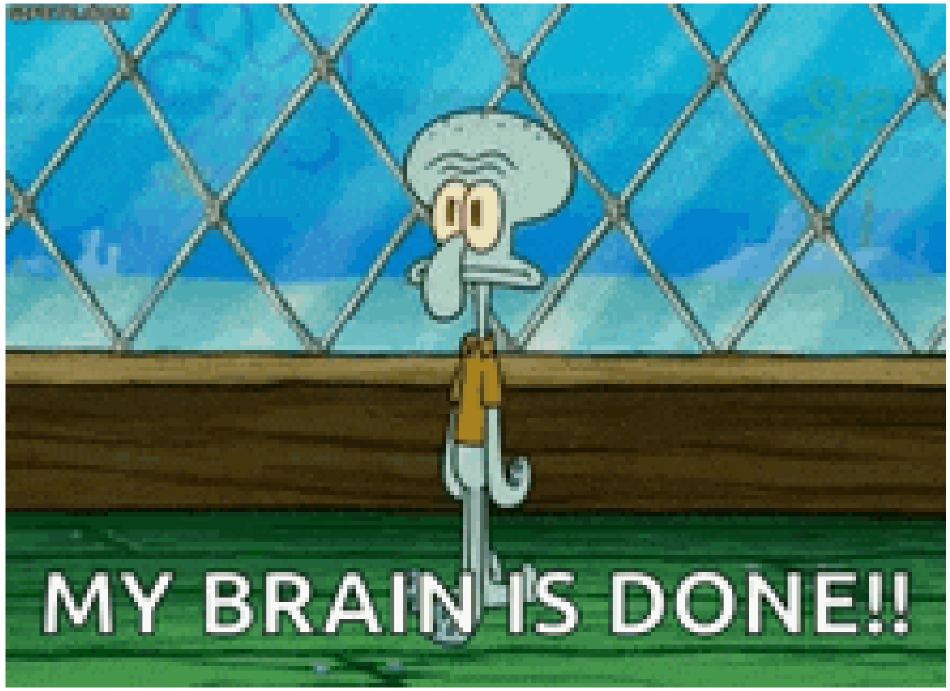
Functions

An advanced subject: functions



Writing your own functions

So far we've seen many functions, like `c()`, `class()`, `filter()`, `dim()` …

Why create your own functions?

- Cut down on repetitive code (easier to fix things!)
- Organize code into manageable chunks
- Avoid running code unintentionally
- Use names that make sense to you

A practical example: summarization

There may be code that you use multiple times. Creating a function can help cut down on repetitive code (and the chance for copy/paste errors).

```
data_insights <- function(x, column1, column2) {
    x_insight <- x %>%
      group_by({{column1}}) %>%
      summarize(mean = mean({{column2}}, na.rm = TRUE))
    return(x_insight)
}

data_insights(x = mtcars, column1 = cyl, column2 = hp)
```

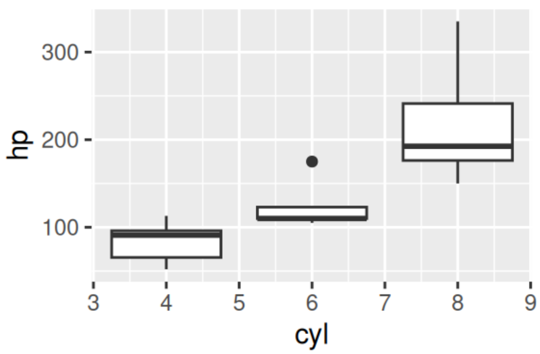
```
# A tibble: 3 × 2
    cyl  mean
  <dbl> <dbl>
1     4  82.6
2     6 122.
3     8 209.
```

A practical example: plotting

You may have a similar plot that you want to examine across columns of data.

```
simple_plots <- function(x, column1, column2) {
    box_plot <- ggplot(data = x, aes(x = {{column1}}, y = {{column2}}, group = {{column1}})) +
      geom_boxplot()
    return(box_plot)
}

simple_plots(x = mtcars, column1 = cyl, column2 = hp)
```



Writing your own functions

The general syntax for a function is:

```
function_name <- function(arg1, arg2, ...) {
 <function body>
}
```

Writing your own functions

Here we will write a function that multiplies some number `x` by 2:

```
times_2 <- function(x) x * 2
```

When you run the line of code above, you make it ready to use (no output yet!). Let's test it!

```
times_2(x = 10)
```

```
[1] 20
```

Writing your own functions: `{ }`

Adding the curly brackets - `{}` - allows you to use functions spanning multiple lines:

```
times_2 <- function(x) {
  x * 2
}
times_2(x = 10)
```

```
[1] 20
```

```
is_even <- function(x) {
  x %% 2 == 0
}
is_even(x = 11)
```

```
[1] FALSE
```

```
is_even(x = times_2(x = 10))
```

```
[1] TRUE
```

Writing your own functions: **return**

If we want something specific for the function's output, we use `return()`:

```
times_2_plus_4 <- function(x) {
  output_int <- x * 2
  output <- output_int + 4
  return(output)
}
times_2_plus_4(x = 10)
```

```
[1] 24
```

Writing your own functions: print intermediate steps

- printed results do not stay around but can show what a function is doing
- returned results stay around
- can only return one result but can print many
- if `return` not called, last evaluated expression is returned
- `return` should be the last step (steps after may be skipped)

Adding print

```
times_2_plus_4 <- function(x) {
  output_int <- x * 2
  output <- output_int + 4
  print(paste("times2 result = ", output_int))
  return(output)
}

result <- times_2_plus_4(x = 10)
```

```
[1] "times2 result =  20"
```

```
result
```

```
[1] 24
```

Writing your own functions: multiple inputs

Functions can take multiple inputs:

```
times_2_plus_y <- function(x, y) x * 2 + y
times_2_plus_y(x = 10, y = 3)
```

```
[1] 23
```

Writing your own functions: multiple outputs

Functions can have one returned result with multiple outputs.

```
x_and_y_plus_2 <- function(x, y) {
  output1 <- x + 2
  output2 <- y + 2

  return(c(output1, output2))
}
result <- x_and_y_plus_2(x = 10, y = 3)
result
```

```
[1] 12  5
```

Writing your own functions: defaults

Functions can have “default” arguments. This lets us use the function without using an argument later:

```
times_2_plus_y <- function(x = 10, y = 3) x * 2 + y
times_2_plus_y()
```

```
[1] 23
```

```
times_2_plus_y(x = 11, y = 4)
```

```
[1] 26
```

Writing another simple function

Let's write a function, `sqdif`, that:

1. takes two numbers `x` and `y` with default values of 2 and 3.
2. takes the difference
3. squares this difference
4. then returns the final value

Writing your own functions: characters

Functions can have any kind of input. Here is a function with characters:

```
loud <- function(word) {
  output <- rep(toupper(word), 5)
  return(output)
}
loud(word = "hooray!")
```

```
[1] "HOORAY!" "HOORAY!" "HOORAY!" "HOORAY!" "HOORAY!"
```

Functions for tibbles

`select(n)` will choose column `n`:

```
get_index <- function(dat, row, col) {
  dat %>%
    filter(row_number() == row) %>%
    select(all_of(col))
}

get_index(dat = iris, row = 10, col = 5)
```

```
  Species
1  setosa
```

Functions for tibbles

Including default values for arguments:

```
get_top <- function(dat, row = 1, col = 1) {
  dat %>%
    filter(row_number() == row) %>%
    select(all_of(col))
}

get_top(dat = iris)
```

```
  Sepal.Length
1          5.1
```

Functions for tibbles - curly braces

Can create function with an argument that allows inputting a column name for `select` or other `dplyr` operation:

```
clean_dataset <- function(dataset, col_name) {
  my_data_out <- dataset %>% select({{col_name}}) # Note the curly braces {{}}
  write_csv(my_data_out, "clean_data.csv")
  return(my_data_out)
}

clean_dataset(dataset = mtcars, col_name = "cyl")
```

```
                    cyl
Mazda RX4             6
Mazda RX4 Wag         6
Datsun 710            4
Hornet 4 Drive        6
Hornet Sportabout     8
Valiant               6
Duster 360            8
Merc 240D             4
Merc 230              4
Merc 280              6
Merc 280C             6
Merc 450SE            8
Merc 450SL            8
Merc 450SLC           8
Cadillac Fleetwood    8
Lincoln Continental   8
Chrysler Imperial     8
```

Functions for tibbles - curly braces

```
# Another example: get means and missing for a specific column
get_summary <- function(dataset, col_name) {
    dataset %>%
    summarise(mean = mean({{col_name}}, na.rm = TRUE),
              na_count = sum(is.na({{col_name}})))
}

get_summary(mtcars, hp)
```

```
      mean na_count
1 146.6875        0
```

Summary

- Simple functions take the form:
  - `NEW_FUNCTION <- function(x, y){x + y}`
  - Can specify defaults like `function(x = 1, y = 2){x + y}` -`return` will provide a value as output
  - `print` will simply print the value on the screen but not save it
- Specify a column (from a tibble) inside a function using `{{double curly braces}}`

Lab Part 1

- Class Website
- Lab

Functions on multiple columns

Using your custom functions: `sapply()`- a base R function

Now that you've made a function… you can "apply" functions easily with `sapply()`!

These functions take the form:

```
sapply(<a vector, list, data frame>, some_function)
```

Using your custom functions: **`sapply()`**

 There are no parentheses on the functions!

You can also pipe into your function.

```
head(iris, n = 2)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
```

```
sapply(iris, class)
```

```
Sepal.Length  Sepal.Width Petal.Length  Petal.Width      Species
   "numeric"    "numeric"    "numeric"    "numeric"     "factor"
```

```
iris %>% sapply(class)
```

```
Sepal.Length  Sepal.Width Petal.Length  Petal.Width      Species
   "numeric"    "numeric"    "numeric"    "numeric"     "factor"
```

Using your custom functions: **sapply()**

```
cars <- read_csv("https://jhudatascience.org/intro_to_r/data/kaggleCarAuction.csv")
select(cars, VehYear:VehicleAge) %>% head()
```

```
# A tibble: 6 × 2
  VehYear VehicleAge
    <dbl>      <dbl>
1    2006          3
2    2004          5
3    2005          4
4    2004          5
5    2005          4
6    2004          5
```

```
select(cars, VehYear:VehicleAge) %>%
  sapply(times_2) %>%
  head()
```

```
     VehYear VehicleAge
[1,]    4012          6
[2,]    4008         10
[3,]    4010          8
[4,]    4008         10
[5,]    4010          8
[6,]    4008         10
```

Using your custom functions “on the fly” to iterate

Also called an “anonymous function”.

```
select(cars, VehYear:VehicleAge) %>%
  sapply(function(x) x / 1000) %>%
  head()
```

```
     VehYear VehicleAge
[1,]   2.006      0.003
[2,]   2.004      0.005
[3,]   2.005      0.004
[4,]   2.004      0.005
[5,]   2.005      0.004
[6,]   2.004      0.005
```

Anonymous functions: alternative syntax

```r
select(cars, VehYear:VehicleAge) %>%
  sapply(\(x) x / 1000) %>%
  head()
```

```
     VehYear VehicleAge
[1,]   2.006      0.003
[2,]   2.004      0.005
[3,]   2.005      0.004
[4,]   2.004      0.005
[5,]   2.005      0.004
[6,]   2.004      0.005
```

across

Using functions in `mutate()` and `summarize()`

Already know how to use functions to modify columns using `mutate()` or calculate summary statistics using `summarize()`.

```
cars %>%
  mutate(VehOdo_round = round(VehOdo, -3)) %>%
  summarize(max_Odo_round = max(VehOdo_round),
            max_Odo = max(VehOdo))
```

```
# A tibble: 1 × 2
  max_Odo_round max_Odo
          <dbl>   <dbl>
1        116000  115717
```

The **across()** function

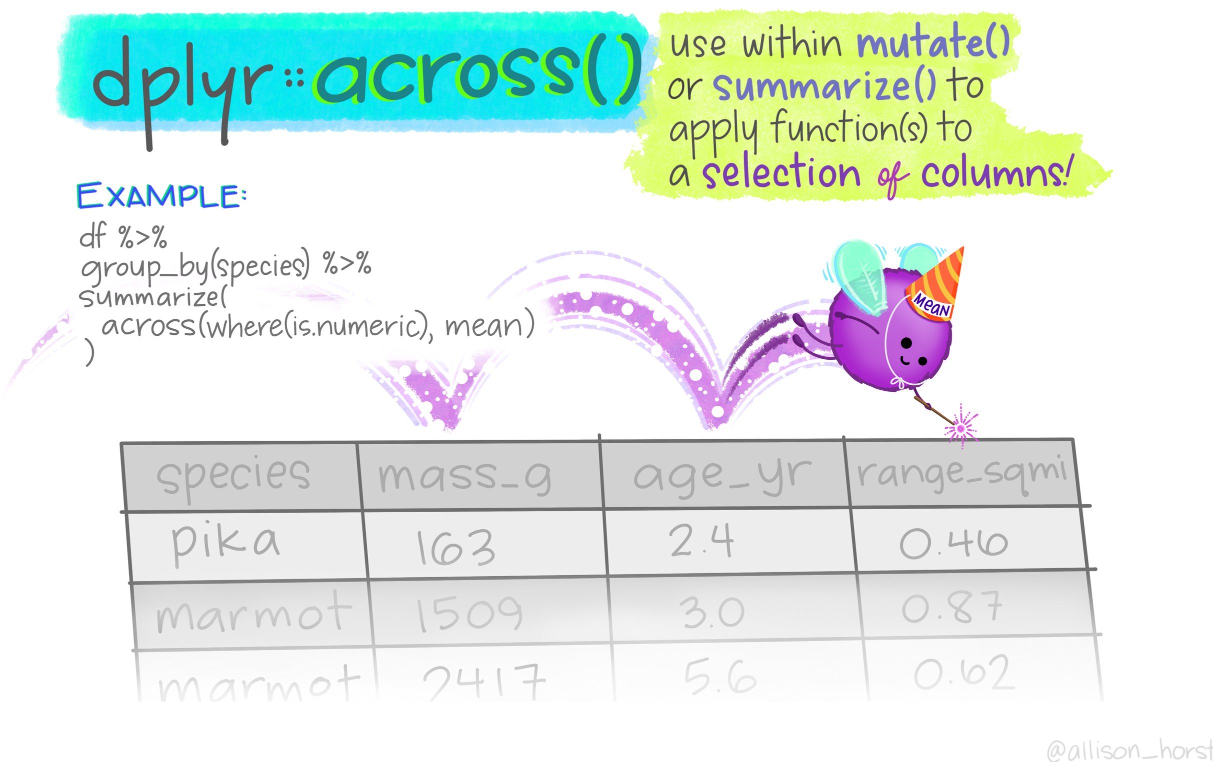


Image by Allison Horst.

Applying functions with `across` from `dplyr`

`across()` makes it easy to apply the same transformation to multiple columns. Usually used with `summarize()` or `mutate()`.

```
summarize(across(.cols = <columns>, .fns = function))
summarize(across(.cols = <columns>, .fns = function(x) {some function}(x, {additional arguments})))
```

or

```
mutate(across(.cols = <columns>, .fns = function))
mutate(across(.cols = <columns>, .fns = function(x) {some function}(x, {additional arguments})))
```

- List columns first: `.cols =`
- List function next: `.fns =`
- If there are arguments to a function (e.g., `na.rm = TRUE`), the function may need to be modified to an anonymous function, e.g., `\(x) mean(x, na.rm = TRUE)`

Applying functions with **across** from **dplyr**

Combining with `summarize()`

```
cars_dbl <- cars %>% select(Make, starts_with("Veh"))

cars_dbl %>%
  summarize(across(.cols = everything(), .fns = mean)) # no parentheses

# A tibble: 1 × 5
   Make VehYear VehicleAge VehOdo VehBCost
  <dbl>   <dbl>      <dbl>  <dbl>    <dbl>
1    NA   2005.       4.18 71500.    6731.
```

Applying functions with **across** from **dplyr**

Can use with other tidyverse functions like `group_by`!

```
cars_dbl %>%
  group_by(Make) %>%
  summarize(across(.cols = everything(), .fns = mean)) # no parentheses
```

```
# A tibble: 33 × 5
   Make      VehYear VehicleAge VehOdo VehBCost
   <chr>       <dbl>      <dbl>  <dbl>    <dbl>
 1 ACURA       2003.       6.52 81732.    9039.
 2 BUICK       2004.       5.65 76238.    6169.
 3 CADILLAC    2004.       5.24 73770.   10958.
 4 CHEVROLET   2006.       3.97 73390.    6835.
 5 CHRYSLER    2006.       3.65 66814.    6507.
 6 DODGE       2006.       3.75 68261.    7047.
 7 FORD        2005.       4.75 76749.    6403.
 8 GMC         2004.       5.61 79273.    8342.
 9 HONDA       2004.       5.33 77877.    8350.
10 HUMMER      2006        3    70809    11920
# ℹ 23 more rows
```

Applying functions with **across** from **dplyr**

To add arguments to functions, may need to use anonymous function. In this syntax, the shorthand `\(x)` is equivalent to `function(x)`.

```
cars_dbl %>%
  group_by(Make) %>%
  summarize(across(.cols = everything(), .fns = \(x) mean(x, na.rm = TRUE)))
```

```
# A tibble: 33 × 5
   Make      VehYear VehicleAge VehOdo VehBCost
   <chr>       <dbl>      <dbl>  <dbl>    <dbl>
 1 ACURA       2003.       6.52 81732.    9039.
 2 BUICK       2004.       5.65 76238.    6169.
 3 CADILLAC    2004.       5.24 73770.   10958.
 4 CHEVROLET   2006.       3.97 73390.    6835.
 5 CHRYSLER    2006.       3.65 66814.    6507.
 6 DODGE       2006.       3.75 68261.    7047.
 7 FORD        2005.       4.75 76749.    6403.
 8 GMC         2004.       5.61 79273.    8342.
 9 HONDA       2004.       5.33 77877.    8350.
10 HUMMER      2006        3    70809    11920
# ℹ 23 more rows
```

Applying functions with **across** from **dplyr**

Using different `tidyselect()` options (e.g., `starts_with()`, `ends_with()`, `contains()`)

```
cars_dbl %>%
  group_by(Make) %>%
  summarize(across(.cols = starts_with("Veh"), .fns = mean))
```

```
# A tibble: 33 × 5
   Make      VehYear VehicleAge VehOdo VehBCost
   <chr>       <dbl>      <dbl>  <dbl>    <dbl>
 1 ACURA       2003.       6.52 81732.    9039.
 2 BUICK       2004.       5.65 76238.    6169.
 3 CADILLAC    2004.       5.24 73770.   10958.
 4 CHEVROLET   2006.       3.97 73390.    6835.
 5 CHRYSLER    2006.       3.65 66814.    6507.
 6 DODGE       2006.       3.75 68261.    7047.
 7 FORD        2005.       4.75 76749.    6403.
 8 GMC         2004.       5.61 79273.    8342.
 9 HONDA       2004.       5.33 77877.    8350.
10 HUMMER      2006        3    70809    11920
# ℹ 23 more rows
```

Applying functions with **across** from **dplyr**

Combining with `mutate()`: rounding to the nearest power of 10 (with negative digits value). Remember that when you need to add additional arguments to a function (e.g., `na.rm`, `digits`, etc.), you must specify an anonymous function with `function(x)` or `\(x)`.

```
cars_dbl %>%
  mutate(across(
    .cols = starts_with("Veh"),
    .fns = function(x) round(x, digits = -3)
  ))
```

```
# A tibble: 72,983 × 5
   Make       VehYear VehicleAge VehOdo VehBCost
   <chr>        <dbl>      <dbl>  <dbl>    <dbl>
 1 MAZDA         2000          0  89000     7000
 2 DODGE         2000          0  94000     8000
 3 DODGE         2000          0  74000     5000
 4 DODGE         2000          0  66000     4000
 5 FORD          2000          0  69000     4000
 6 MITSUBISHI    2000          0  81000     6000
 7 KIA           2000          0  65000     4000
 8 FORD          2000          0  66000     4000
 9 KIA           2000          0  50000     6000
10 FORD          2000          0  85000     8000
# ℹ 72,973 more rows
```

Applying functions with **across** from **dplyr**

Combining with `mutate()` - the `replace_na` function

`replace_na({data frame}, {list of values})` or `replace_na({vector}, {single value})`

```
# Child mortality data
mort <-
  read_csv("https://jhudatascience.org/intro_to_r/data/mortality.csv") %>%
  rename(country = `...1`)

mort %>%
  select(country, starts_with("194")) %>%
  mutate(across(
    .cols = c(`1943`, `1944`, `1945`),
    .fns = function(x) replace_na(x, replace = 0)
  ))
```

```
# A tibble: 197 × 11
   country `1940` `1941` `1942` `1943` `1944` `1945` `1946` `1947` `1948` `1949`
   <chr>    <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
 1 Afghan… NA     NA     NA      0      0      0     NA     NA     NA     NA
 2 Albania  1.53   1.31   1.48   1.46   1.43   1.40   1.37   1.41   1.37   1.34
 3 Algeria NA     NA     NA      0      0      0     NA     NA     NA     NA
 4 Angola   4.46   4.46   4.46   4.34   4.34   4.34   4.33   4.22   4.22   4.21
 5 Argent…  0.641  0.603  0.602  0.558  0.551  0.510  0.503  0.496  0.494  0.492
 6 Armenia NA     NA     NA      0      0      0     NA     NA     NA     NA
 7 Aruba   NA     NA     NA      0      0      0     NA     NA     NA     NA
 8 Austra…  0.263  0.275  0.276  0.299  0.260  0.271  0.295  0.279  0.271  0.271
 9 Austria  0.504  0.474  0.417  0.389  0.360  0.311  0.311  0.312  0.274  0.274
10 Azerba… NA     NA     NA      0      0      0     NA     NA     NA     NA
# ℹ 187 more rows
```

Use custom functions within **mutate** and **across**

If your function needs to span more than one line, better to define it first before using inside `mutate()` and `across()`.

```
times1000 <- function(x) x * 1000

airquality %>%
  mutate(across(
    .cols = everything(),
    .fns  = times1000
  )) %>%
  head(n = 2)

  Ozone Solar.R Wind  Temp Month  Day
1 41000  190000 7400 67000  5000 1000
2 36000  118000 8000 72000  5000 2000

airquality %>%
  mutate(across(
    .cols = everything(),
    .fns  = function(x) x * 1000
  )) %>%
  head(n = 2)

  Ozone Solar.R Wind  Temp Month  Day
1 41000  190000 7400 67000  5000 1000
2 36000  118000 8000 72000  5000 2000
```

GUT CHECK!

Why use `across()`?

A. Efficiency - faster and less repetitive

B. Calculate the cross product

C. Connect across datasets

`purrr` package

Similar to across, `purrr` is a package that allows you to apply a function to multiple columns in a data frame or multiple data objects in a list.

A *list* in R is a generic class of data consisting of an ordered collection of objects. It can include any number of single numeric objects, vectors, or data frames – can be all the same class of objects or all different.

While we won't get into `purrr` too much in this class, its a handy package for you to know about should you get into a situation where you have an irregular list you need to handle!

Multiple Data Frames

Multiple data frames

Lists help us work with multiple data frames

```
AQ_list <- list(AQ1 = airquality, AQ2 = airquality, AQ3 = airquality)
str(AQ_list)
```

```
List of 3
 $ AQ1:'data.frame':	153 obs. of  6 variables:
  ..$ Ozone  : int [1:153] 41 36 12 18 NA 28 23 19 8 NA ...
  ..$ Solar.R: int [1:153] 190 118 149 313 NA NA 299 99 19 194 ...
  ..$ Wind   : num [1:153] 7.4 8 12.6 11.5 14.3 14.9 8.6 13.8 20.1 8.6 ...
  ..$ Temp   : int [1:153] 67 72 74 62 56 66 65 59 61 69 ...
  ..$ Month  : int [1:153] 5 5 5 5 5 5 5 5 5 5 ...
  ..$ Day    : int [1:153] 1 2 3 4 5 6 7 8 9 10 ...
 $ AQ2:'data.frame':	153 obs. of  6 variables:
  ..$ Ozone  : int [1:153] 41 36 12 18 NA 28 23 19 8 NA ...
  ..$ Solar.R: int [1:153] 190 118 149 313 NA NA 299 99 19 194 ...
  ..$ Wind   : num [1:153] 7.4 8 12.6 11.5 14.3 14.9 8.6 13.8 20.1 8.6 ...
  ..$ Temp   : int [1:153] 67 72 74 62 56 66 65 59 61 69 ...
  ..$ Month  : int [1:153] 5 5 5 5 5 5 5 5 5 5 ...
  ..$ Day    : int [1:153] 1 2 3 4 5 6 7 8 9 10 ...
 $ AQ3:'data.frame':	153 obs. of  6 variables:
  ..$ Ozone  : int [1:153] 41 36 12 18 NA 28 23 19 8 NA ...
  ..$ Solar.R: int [1:153] 190 118 149 313 NA NA 299 99 19 194 ...
  ..$ Wind   : num [1:153] 7.4 8 12.6 11.5 14.3 14.9 8.6 13.8 20.1 8.6 ...
  ..$ Temp   : int [1:153] 67 72 74 62 56 66 65 59 61 69 ...
  ..$ Month  : int [1:153] 5 5 5 5 5 5 5 5 5 5 ...
  ..$ Day    : int [1:153] 1 2 3 4 5 6 7 8 9 10 ...
```

Multiple data frames: **sapply**

```
AQ_list %>% sapply(class)
```

```
         AQ1          AQ2          AQ3
"data.frame" "data.frame" "data.frame"
```

```
AQ_list %>% sapply(nrow)
```

```
AQ1 AQ2 AQ3
153 153 153
```

```
AQ_list %>% sapply(colMeans, na.rm = TRUE)
```

```
               AQ1        AQ2        AQ3
Ozone    42.129310  42.129310  42.129310
Solar.R 185.931507 185.931507 185.931507
Wind      9.957516   9.957516   9.957516
Temp     77.882353  77.882353  77.882353
Month     6.993464   6.993464   6.993464
Day      15.803922  15.803922  15.803922
```

Summary

- Apply your functions with `sapply(<a vector or list>, some_function)`
- Use `across()` to apply functions across multiple columns of data
- Need to use `across` within `summarize()` or `mutate()`
- Can use `sapply` or `purrr` to work with multiple data frames within lists simultaneously

Lab Part 2

- Class Website
- Lab
- Day 9 Cheatsheet
- Posit's `purrr` Cheatsheet

Image by Gerd Altmann from Pixabay

Good luck and happy coding!

Image by Allison Horst.